Reproduction in organisms

All living things are made up of *cells*. Cells are the building blocks of life. The simplest forms of life are *organisms* made up of only one cell. All organisms must reproduce in order to make up for losses due to death. If more organisms are being created than are dying at any one time there is pressure for the organism to spread and colonise new *habitats*.

1 What is the general term for an organism consisting of one cell?

2 Cells are usually described as *microscopic*. What does this mean?

3 Give a definition for *habitat*.

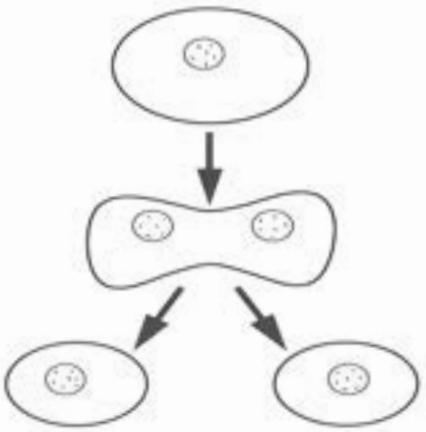
4 Describe what is likely to happen to organisms that stay in one area which has only limited resources.

Organisms reproduce by two means: *asexual* reproduction and *sexual* reproduction.

Asexual reproduction

Asexual (meaning 'without sex') is the simplest form of reproduction. There are a number of different types but all share two main features: only one parent is needed and the new individual produced is always genetically identical to this parent.

Some simple creatures (eg amoeba) split into two daughter cells by binary fission. Others (eg yeast and Hydra) form groups of cells that grow out of the organism and develop into a new individual by a process called budding.



5 The diagram shows a simplified version of how binary fission occurs. Use the diagram to describe the process.

6 A particular *bacterium* (plural *bacteria*) reproduces by binary fission. If this division occurs, on average, once every 20 minutes, calculate how many bacteria theoretically could exist after 12 hours starting from a single bacterium.

7 In real life, would you expect the number of bacteria calculated in question 6 to arise? Explain your answer.

Fungi, ferns and mosses produce spores that break off the parent when mature. They are carried about by the wind, by rivers or by sticking to the bodies of animals. Each and every spore is capable of developing into a new individual, provided that it lands in suitable conditions. Even if spores do not find such conditions immediately, they are often capable of prolonged survival until conditions change for the better. Organisms that produce spores create large numbers of them. Since close to 100% of these spores could produce new individuals, this method of reproduction represents great economy of material and effort. However, each spore produced by the organism carries the characteristics of its single parent and will develop into a replica of that parent. There is no variation.

8 Identify possible suitable conditions for a spore.

9 Name two organisms that can produce spores.

10 Give the definition of *offspring*.

11 Name two ways that can be used to spread spores.

12 Name two types of organism that reproduce by budding.

Plants can also be *propagated*, which involves taking a cutting from one plant and growing that into other (identical) plants.

13 What is the name for an organism that is genetically identical to its parent?

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Sexual reproduction

Sexual reproduction involves two parent sex cells, which produce sex cells (gametes). For example:

	Female cell	Male cell
Animals	egg (ovum)	sperm
Plants	ovule	pollen

One male cell fertilises one female cell to produce a *zygote*, which is the first cell of a new living organism. The offspring formed by this combination of sex cells possess a mixture of the characteristics of the two parents. For this reason they are always slightly different to both parents. Because of this we get *variation* and variation has been shown to make a species stronger and more able to overcome competitors, survive hard times such as drought, climatic change and other natural disasters.

14 How many parents are involved in asexual reproduction?

15 How many parents are involved in sexual reproduction?

16 Outline why variation of individuals within a species might help that species survive.

Sexual reproduction has two major disadvantages. First, it is slow. The formation of sex cells is a lengthy business and the female cell must wait to be fertilised by male cells. Male cells must find the female cell. Secondly, sexual reproduction is wasteful. All sex cells represent valuable material and effort from the parents. The vast majority of male cells produced fail to find a female cell. They are often spread around at random. For example, the pollen of grass plants is distributed by the wind; 90% of this pollen fails to land on the flowers of a plant of the same species. Because of this many plants and animals have to produce sex cells in large numbers.

17 Identify one advantage and one disadvantage of sexual reproduction.

18 What is a *zygote*?

19 What are *gametes*?

20 Why are organisms formed by sexual reproduction not identical?

21 Why do plants and animals often have to produce so many sex cells?

Organisms in which fertilisation and development of the embryo take place inside the body are called *viviparous* and the baby is born live. In *oviparous* organisms, the development of an embryo occurs inside an egg that has been laid by the mother.



22 Name three viviparous organisms.

23 Name three oviparous organisms.

24 Define *embryo*.

25 Show how the words *viviparous* and *oviparous* are derived. You may need to use a dictionary.

26 Give two disadvantages of asexual reproduction.

27 Give two examples of natural events that might endanger a species.

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